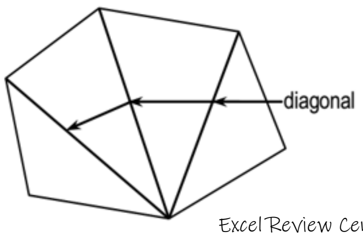
Plane Geometry 3

Diagonals of Polygons

The diagonal of a polygon is a line connecting two opposite vertices.



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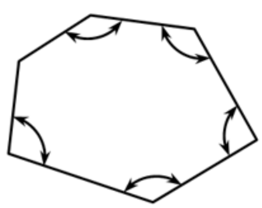
The number of diagonals of polygon of n sides is:

$$\text{diagonals} = \frac{n}{2}(n-3)$$

Where:
n = number of sides of polygon

Sum of interior angles

The interior angles of a polygon are the angles inside the polygon.

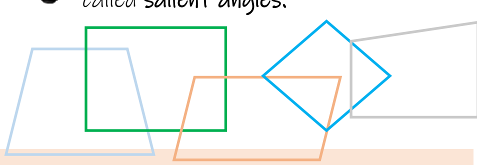


The sum of interior angles of polygon of n sides is:

$$\sum \text{int angles} = (n-2)(180^\circ)$$

Where:
n = number of sides of polygon

REENTRANT ANGLE – is the inward-pointing angle of the concave polygon while the other angles are called **salient angles**.



Quadrilaterals

A quadrilateral is a polygon with four sides. A quadrilateral is also known as quadrangle or tetragon.

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Types of quadrilaterals

1. **Square** – a regular quadrilateral. All its sides are equal and all angles are equal to right angle. A square is a rectangle with all sides equal.
2. Rectangle- a right-angled parallelogram
3. **Rhombus** – all sides are equal but no angle equal to right angle
4. **Parallelogram** – both pairs of opposite sides are parallel. Another term for parallelogram is rhomboid.
5. **Trapezoid** – only two sides are parallel
6. **Trapezium** – if no two sides are parallel
7. **Kite** – convex quadrilateral who adjacent sides are equal in pair.
8. **Deltoid** – concave quadrilateral whose adjacent sides are equal in pair
9. **Cyclic quadrilateral** – a quadrilateral whose vertices lie on a circle.

Formulas:

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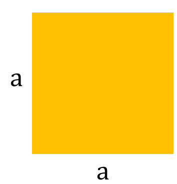
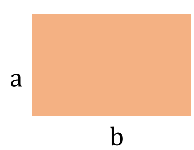
1. Square



$$\text{Area} = a^2$$

$$\text{Perimeter} = 4a$$

2. Rectangle

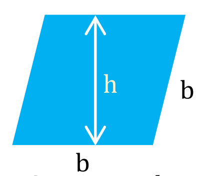


$$\text{Area} = ab$$

$$\text{Perimeter} = 2(a+b)$$

3. Rhombus

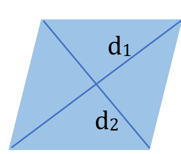
A. Given base and height



$$\text{Area} = bh$$

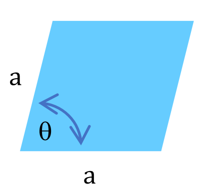
$$\text{Perimeter} = 4b$$

B. Given two diagonals



$$\text{Area} = \frac{1}{2}d_1 d_2$$

C. Given two sides and included angle

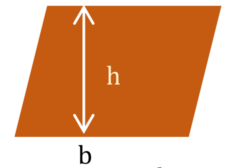


$$\text{Area} = a^2 \sin\theta$$

NOTE:
The angle between two diagonals of a rhombus is equal to 90° or right angle

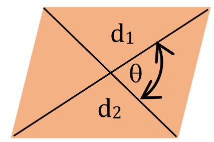
4. Parallelogram

A. Given base and height



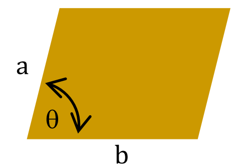
$$\text{Area} = bh$$

B. Given two diagonals and their included angle



$$\text{Area} = \frac{1}{2}d_1 d_2 \sin\theta$$

C. Given two sides and included angle



$$\text{Area} = ab\sin\theta$$

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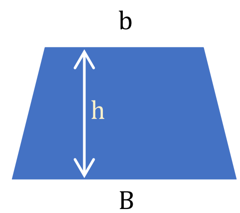
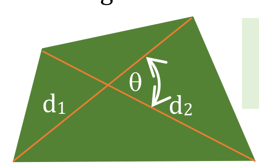
5. Trapezoid



$$\text{Area} = \frac{1}{2}(b+B)h$$

6. Trapezium (General quadrilateral)

A. Given two diagonals and included angle



$$\text{Area} = \frac{1}{2}d_1 d_2 \sin\theta$$

A. Given four sides and opposite angles

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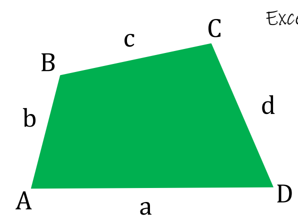

Use Brahmagupta's formula

$$A = \sqrt{(s-a)(s-b)(s-c)(s-d) - abcd\cos^2\theta}$$

Where:

$$\theta = \frac{A+C}{2} = \frac{B+D}{2} \qquad s = \frac{a+b+c+d}{2}$$

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